

Pilot Assignment in Cell-Free Massive MIMO

Conflict-Graph Sparsification for Pilot-Overhead Control

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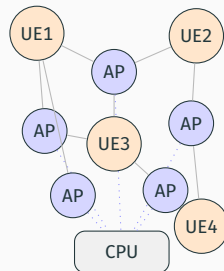
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Basic idea

- Many distributed APs connect to a central processing unit.
- From the UE side, there is no fixed cell boundary.
- A UE can be served by several APs.
- An AP can receive signals from several UEs.

Implication

- Macro-diversity improves coverage.
- Many AP-UE channel links must be estimated.
- Pilot assignment becomes a network-level problem.



Channel between AP l and UE k

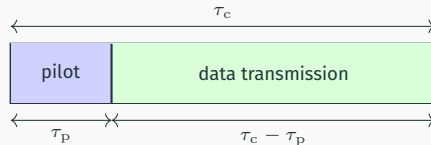
$$\mathbf{h}_{lk} = \sqrt{\beta_{lk}} \mathbf{g}_{lk}, \quad \mathbf{g}_{lk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$$

- L APs, K UEs, N antennas per AP.
- β_{lk} captures pathloss and shadowing.
- Dominant APs/beams are the useful structure for pilot assignment.

Spectral efficiency form

$$\text{SE}_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

Coherence block



Key tension

More pilot symbols can reduce reuse pressure, but they leave fewer symbols for data.

Pilot observation at AP l

$$\begin{aligned}\mathbf{Y}_l^p &= \sum_{i=1}^K \sqrt{p_i} \mathbf{h}_{li} \phi_{t_i}^T + \mathbf{N}_l, \\ \mathbf{y}_{lk}^p &= \mathbf{Y}_l^p \phi_{t_k}^* \\ &= \sqrt{p_k} \tau_p \mathbf{h}_{lk} + \sum_{i \in \mathcal{P}_k \setminus \{k\}} \sqrt{p_i} \tau_p \mathbf{h}_{li} + \mathbf{n}_{lk}. \\ \mathcal{P}_k &= \{i : t_i = t_k\}\end{aligned}$$

Interpretation

- UE k transmits pilot ϕ_{t_k} .
- AP l correlates the received signal with that pilot.
- If another UE uses the same pilot, its channel enters the same statistic.
- The estimate is then contaminated before data detection starts.

Pilot contamination

Same pilot plus overlapping strong channels creates mixed channel estimates.

Why assignment is difficult

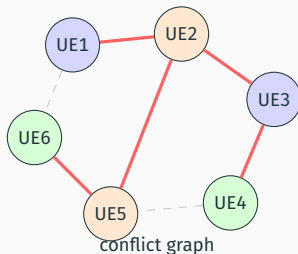
- Orthogonal pilot count is limited: usually $\tau_p < K$.
- Pilot reuse is therefore necessary.
- Each UE can overlap with many other UEs across APs and beams.
- A high-overlap same-pilot pair lowers channel-estimation quality and SINR.

Graph view

- Node = UE.
- Edge = strong effective conflict.
- Color = pilot index.

Research question

Which UEs can reuse a pilot without creating a strong effective conflict?



fewer colors mean fewer actual pilots

$$SE_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

More pilots

- More orthogonal pilot resources.
- Lower pilot reuse pressure.
- Higher training overhead.
- Less data time in the coherence block.

Fewer pilots

- Larger prelog factor.
- Better data-time efficiency.
- Higher contamination risk.
- Works only if the retained structure keeps SINR strong.

Main lens

A useful assignment should reduce pilot overhead without throwing away too much channel quality.

Gao et al.

- AP-domain pilot assignment.
- Many-to-many matching idea.
- Strong small-pilot regime result.
- Pilot count is a fixed design input.

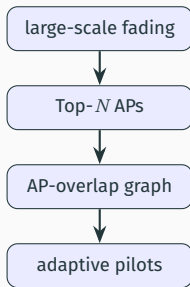
Mussbah et al.

- Beam-domain active/moderate beam graph.
- Conflict graph colored by DSATUR.
- Actual pilot count can become adaptive.
- RF-chain activity matters for EE.

Family	Domain	Conflict signal	Pilot-count rule
Gao	AP / large-scale fading	UE-AP matching	fixed design budget
Mussbah	beam domain	active/moderate beam overlap	graph chromatic number
Proposed set	AP + beam variants	sparsified effective conflicts	adaptive where possible

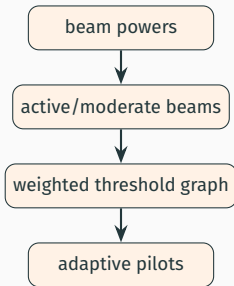
AP-Top-N Adaptive

- Strongest N_{top} APs per UE.
- Conflict by shared Top- N APs.
- Adaptive graph coloring.



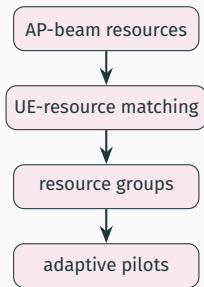
Beam-Weighted Threshold

- Active/moderate beam split.
- Higher weight for active-active overlap.
- Threshold weak conflicts.



Beam-Resource Matching

- Candidate AP-beam resources.
- Many-to-many UE-resource matching.
- Adaptive pilot grouping.



Method	Domain	Conflict signal	Pilot count	Role
Random	AP	none	fixed	reference
Gao Matching	AP	UE-AP matching	fixed	reference
Mussbah Beam Graph	beam	binary beam overlap	adaptive	reference
AP-Top-N Adaptive	AP	Top- N AP overlap	adaptive	proposed
Beam-Weighted Threshold	beam	weighted beam overlap	adaptive	proposed
Beam-Resource Matching	beam	AP-beam resource matching	adaptive	proposed

Unifying view

Different conflict signals, same lever: reduce effective conflicts while keeping dominant channel structure.

Common simulation setting

Parameter	Value
APs	$L = 200$
AP antennas	$N = 8$ unless swept
Coherence block	$\tau_c = 150$
Fixed pilot budget	$\tau_p^{design} = 15$
Monte Carlo setups	200
Power control	full
Carrier frequency	3 GHz

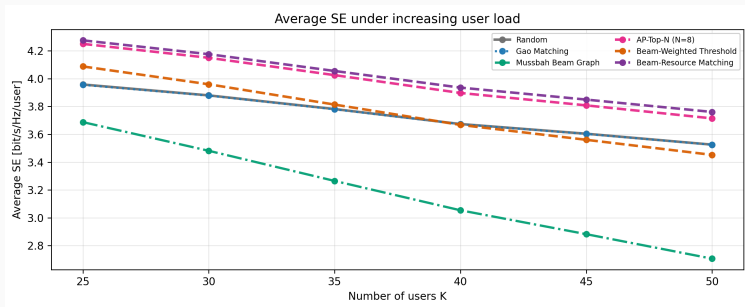
Adaptive methods use their actual graph-coloring pilot count in the SE prelog.

Methods compared

- Random
- Gao Matching
- Mussbah Beam Graph
- AP-Top-N Adaptive
- Beam-Weighted Threshold
- Beam-Resource Matching

Main evaluation question

Does the conflict graph reduce pilot overhead while keeping enough useful channel structure?



Environment: $K = 25, \dots, 50$, $L = 200$, $N = 8$, $\tau_c = 150$,
 $\tau_p^{design} = 15$, 200 setups.

Result at $K = 50$

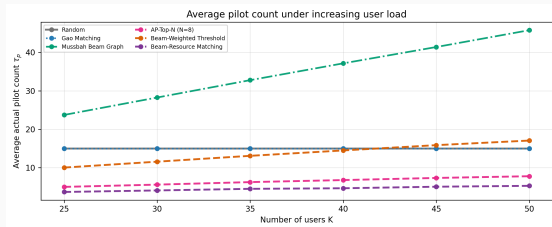
- Beam-Resource: +6.66% mean SE, +7.07% 95%-likely SE.
- AP-Top-N: +5.35% mean SE, +6.32% 95%-likely SE.
- Gao stays near Random: -0.02% mean SE.
- Mussbah loses 23.24% mean SE as actual τ_p grows.

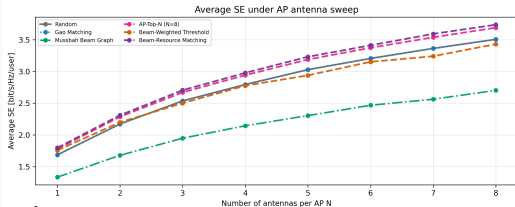
Same-environment K-sweep

$K = 25, \dots, 50$, $L = 200$, $N = 8$, $\tau_c = 150$,
 $\tau_p^{design} = 15$, 200 setups.

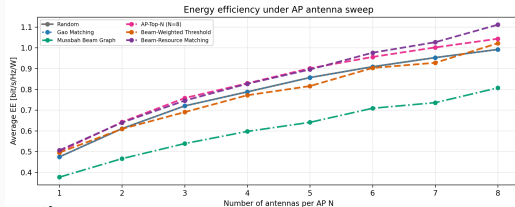
Reading

- Random and Gao stay fixed at $\tau_p = 15$.
- Mussbah grows from 23.7 to 45.8 pilots as load increases.
- AP-Top-N and Beam-Resource stay below the design pilot budget.
- Beam-Weighted crosses the budget under heavier load.



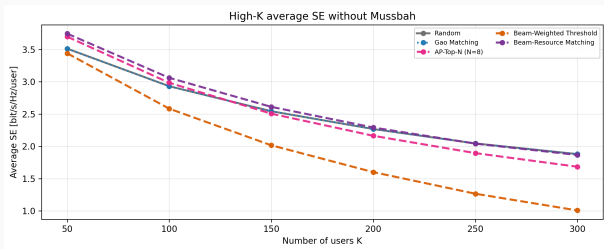


Environment: $K = 50$, $L = 200$, $N = 1, \dots, 8$, $\tau_c = 150$,
 $\tau_p^{design} = 15$, 200 setups.



Reading

- Beam-Resource is the best mean-SE method for every tested N .
- At $N = 8$: Beam-Resource gives +6.57% SE and +12.01% EE proxy.
- AP-Top-N remains positive and simpler: +5.25% SE at $N = 8$.



Why this plot matters

- Only K is increased from the main setting.
- Gao still stays close to Random: +0.07% mean SE at $K = 300$.
- Beam-Resource stays positive through $K = 200$, then crosses below Random.
- AP-Top-N crosses earlier, around $K = 150$.

Plot choice

- Mussbah is omitted because its actual τ_P exceeds τ_c from $K = 200$.

Method	Main observation	Strength	Weakness
Random	fixed-pilot reference	stable baseline	no conflict awareness
Gao Matching	near Random here	useful AP-domain reference	small effect in this evaluator
Mussbah Beam Graph	dense graph, many pilots	beam-domain reference	τ_p grows too fast
AP-Top-N Adaptive	positive moderate-load gain	simple large-scale rule	loses earlier at high K
Beam-Weighted Threshold	exposes dense-graph failure	active-beam weighting idea	pilot count not controlled enough
Beam-Resource Matching	strongest current direction	low τ_p with channel structure	has an operating range

Interpretation

The useful design is not the one that minimizes pilot count alone; it is the one that removes weak conflicts while preserving enough dominant AP/beam structure.

Pilot assignment is a balance problem.
Prune enough to reduce pilots, but not enough to destroy SINR.

What we can defend

- Beam-Resource is the strongest current rule in this evaluator.
- AP-Top-N is a simple AP-domain rule with positive moderate-load gain.
- Beam-Weighted and Mussbah show why pilot-count control matters.

What we should not oversell

- Fewer pilots are not always better.
- Beam-domain information is useful only if the graph is not too dense.
- Gao remains close to Random in the same-environment comparison.
- EE is a simplified proxy.

Final claim: adaptive pilot assignment is useful when it cuts pilot overhead without sacrificing too much useful channel structure.

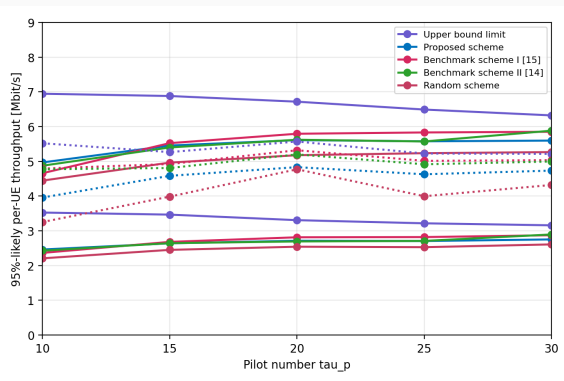
Backup: Gao Small-Pilot Reproduction Anchor

What this supports

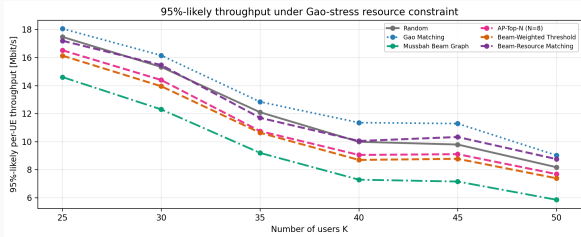
- 200 Monte Carlo setups.
- $\tau_p = 10$, max-min power.
- Gao vs Random: +21.7% in this reproduction run.
- Paper claim: approximately +23%.

How it is used

- Reproduction anchor for Gao-style behavior.
- Not the final same-environment ranking.



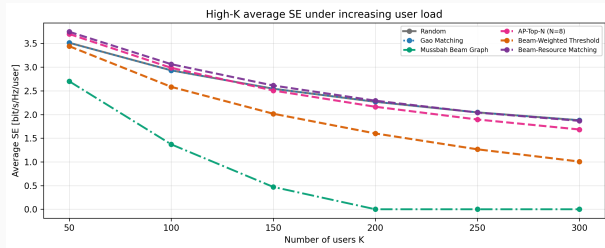
Backup: Gao Small-Pilot Diagnostic



Why this is backup

- $\tau_p = 3$ intentionally makes the pilot budget very tight.
- This helps diagnose when Gao-style matching becomes visible.
- It is not the practical main setting of the final comparison.

Backup: Full High- K Plot with Mussbah



Why the main slide omits Mussbah

- Mussbah actual τ_p : 160.7 at $K = 200$.
- Since $\tau_c = 150$, the SE prelog collapses.
- The resulting zero-SE curve hides differences among the other methods.

Backup: Graph Consistency Rules

- 1 Use same-environment MC figures for the main ranking.
- 2 Use method names, not person-specific labels.
- 3 Use only Random, Gao Matching, and Mussbah Beam Graph as main references.
- 4 Use Gao reproduction only as a small-pilot reproduction anchor.
- 5 Say “EE proxy” for the current EE plots.
- 6 Claim crossover only for the methods that actually cross in the final six-method plot.
- 7 Add bootstrap CI before using significance language.